Introduction to
Internet of Things
Assignment-Week 12

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 15 **Total marks: 15 X 1= 15**

QUESTION 1:

Which type of data analysis is used for data that is categorical in nature?

a. Qualitative analysis

b. Quantitative analysis

c. Regression analysis

d. Analysis of variance

Correct Answer: a. Qualitative analysis

Detailed Solution: Qualitative analysis deals with categorical data, whereas quantitative analysis is used for numerical data.
*See Lecture 56 (Data Handling and Analytics - Part II, **slide 3**)*

QUESTION 2:

According to the lecture, when is a statistical model classified as a complete model?

a. When the number of variables equals the number of equations

b. When the model contains no variables

c. When the number of variables is always greater than the number of equations

d. When only categorical data are used

Correct Answer: a. When the number of variables equals the number of equations

Detailed Solution: A complete statistical model has the same number of variables as equations. If these numbers do not match, the model is incomplete.
*See Lecture 56 (Data Handling and Analytics - Part II, **slide 9**)*

QUESTION 3:

Analysis of Variance (ANOVA) is best applied when which of the following are to be compared?

a. Only a single observation

b. More than two populations or samples

c. Only non-numerical labels without any response variable

d. Only identical copies of the same observation

Correct Answer: b. More than two populations or samples

Detailed Solution: ANOVA is introduced as a parametric statistical technique for comparing datasets and is especially useful when more than two populations or samples are being compared.
*See Lecture 56 (Data Handling and Analytics - Part II, **slide 11**)*

QUESTION 4:

What does Pearson's correlation coefficient (r) measure?

a. The number of equations in a statistical model

b. The maximum value in a dataset

c. The strength of association between two variables

d. The sample size of an experiment

Correct Answer: c. The strength of association between two variables

Detailed Solution: Pearson's correlation coefficient is used to measure the strength of association between two quantitative, continuous variables.
*See Lecture 56 (Data Handling and Analytics - Part II, **slide 18**)*

QUESTION 5:

What is a primary objective of the AgriSens smart water management system?

a. Restricting irrigation control to manual operation only

b. Increasing irrigation water use regardless of soil condition

c. Removing all sensors from agricultural fields

d. Achieving more crop yield with less water

Correct Answer: d. Achieving more crop yield with less water

Detailed Solution: AgriSens aims to improve water management by using less water while maintaining or improving crop yield, along with automatic irrigation and remote monitoring.
*See Lecture 57 (Case Study: Agriculture, **slide 4**)*

QUESTION 6:

Which set correctly represents the three layers of the proposed AgriSens architecture?

a. Physical layer; transport layer; presentation layer

b. Sensing and actuating layer; processing, storage and service layer; application layer

c. User layer; billing layer; routing layer

d. Sensor layer; firewall layer; operating-system layer

Correct Answer: b. Sensing and actuating layer; processing, storage and service layer; application layer

Detailed Solution: The proposed AgriSens architecture is organized into a sensing and actuating layer, a processing, storage and service layer, and an application layer.
*See Lecture 57 (Case Study: Agriculture, **slide 5**)*

QUESTION 7:

Which component is included in the integrated design of an AgriSens sensor node?

a. Meter data management system for smart grids

b. Power management unit

c. Phasor measurement unit

d. Hadoop Namenode

Correct Answer: b. Power management unit

Detailed Solution: The AgriSens sensor node contains sensors and actuators, a processing and memory unit, a wireless communication unit, and a power management unit.
*See Lecture 57 (Case Study: Agriculture, **slide 8**)*

QUESTION 8:

What is the function of the multi-user server in the AgriSens remote-server design?

a. It supplies electrical power to the sensor nodes

b. It measures soil moisture directly in the field

c. It sends field information to the farmer's mobile phone using SMS and executes farmer queries and control messages

d. It replaces all gateways with wired links

Correct Answer: c. It sends field information to the farmer's mobile phone using SMS and executes farmer queries and control messages

Detailed Solution: The multi-user server communicates field information to farmers through SMS and also processes farmer queries and control commands.

*See Lecture 57 (Case Study: Agriculture, **slide 10**)*

QUESTION 9:

In the IoT healthcare architecture discussed in the lecture, which layer contains smartphones and tablets used as data aggregators?

a. Physical power layer

b. Sensing layer

c. Cloud platform

d. Aggregated layer

Correct Answer: d. Aggregated layer

Detailed Solution: The aggregated layer collects data from sensing-layer devices. Smartphones and tablets are given as examples of aggregators.
*See Lecture 58 (Case Study: Healthcare, **slide 3**)*

QUESTION 10:

Which of the following is an example of preventive care enabled by IoT healthcare?

a. Fall detection for senior citizens

b. Disabling emergency alerts

c. Removing physiological sensors from patients

d. Storing only paper-based medical records

Correct Answer: a. Fall detection for senior citizens

Detailed Solution: The lecture presents fall detection, emergency situation detection, and early anomaly detection as examples of preventive care using IoT healthcare.
*See Lecture 58 (Case Study: Healthcare, **slide 8**)*

QUESTION 11:

What is an advantage of continuous activity monitoring?

a. It requires all sensor data to be discarded immediately

b. It prevents any long-term observation

c. It enables daily observation of human behavior and repetitive activity patterns

d. It can operate only with cameras

Correct Answer: c. It enables daily observation of human behavior and repetitive activity patterns

Detailed Solution: Continuous monitoring helps observe daily behavior over long periods and identify repetitive patterns that can improve the reliability of activity recognition.
*See Lecture 59 (Activity Monitoring - Part I, **slide 5**)*

QUESTION 12:

Which type of data can be analyzed using the deep-learning-based approach described in the lecture?

a. Sensor data, images, and videos

b. Images only

c. Text labels only

d. Sensor data only and never images or videos

Correct Answer: a. Sensor data, images, and videos

Detailed Solution: The lecture states that deep-learning-based analysis can be applied to ordinary sensor data as well as image and video data.
*See Lecture 59 (Activity Monitoring - Part I, **slide 8**)*

QUESTION 13:

Processing activity-monitoring sensor data directly on the device is referred to as which approach?

a. Data-aggregator selection approach

b. Network-only approach

c. Cloud replication approach

d. In-place approach

Correct Answer: d. In-place approach

Detailed Solution: In the in-place approach, activity data are processed on the device itself. This is generally power intensive but does not require a network connection for the processing task.
*See Lecture 59 (Activity Monitoring - Part I, **slide 9**)*

QUESTION 14:

What are the three basic elements used in the smartphone-based activity-monitoring demonstration?

a. A soil-moisture sensor, a pump, and a paddy field

b. A smart meter, a PMU, and a power plant

c. A smartphone, a network, and a remote server

d. A Hadoop Namenode, Datanode, and Job Tracker

Correct Answer: c. A smartphone, a network, and a remote server

Detailed Solution: The demonstration uses a smartphone to generate sensor data, a network to carry the data, and a remote server to receive and process or visualize it.
See Lecture 60 (Activity Monitoring - Part II)

QUESTION 15:

Which sensor values are specifically logged in the activity-monitoring demonstration described in the lecture?

a. Accelerometer values

b. Soil-moisture values

c. Smart-grid voltage phasors

d. Water-level values

Correct Answer: a. Accelerometer values

Detailed Solution: For the demonstration, the smartphone transmits accelerometer readings to the remote server, where the data are continuously logged.
See Lecture 60 (Activity Monitoring - Part II)

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